

Question	Answer	Marks
4(a)	graph with x-axis labelled 'distance' and wavelength/ λ correctly shown	B1
	graph with x-axis labelled 'time' and period/ T correctly shown	B1
	graph with y-axis labelled 'displacement' and amplitude/ A correctly shown	B1
4(b)(i)	wave (moves along string and) reflects at fixed point/Y/X/end/wall/boundary	B1
	the incident and reflected waves interfere/superpose	B1
4(b)(ii)	100/40 or 2.5 (cycles/periods/ T)	C1
	1. displacement = 0	B1
	2. distance = 130 mm	A1
4(b)(iii)	1. $f = 1/40 \times 10^{-3}$ = 25 Hz	A1
	2. $v = f\lambda$ or $\lambda = vT$	C1
	$\lambda = 30/25$ or $30 \times 40 \times 10^{-3}$ (= 1.2 m)	C1
	distance = 1.2×1.5 = 1.8 m	A1

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5(c)	$5 = 5/0$	M1